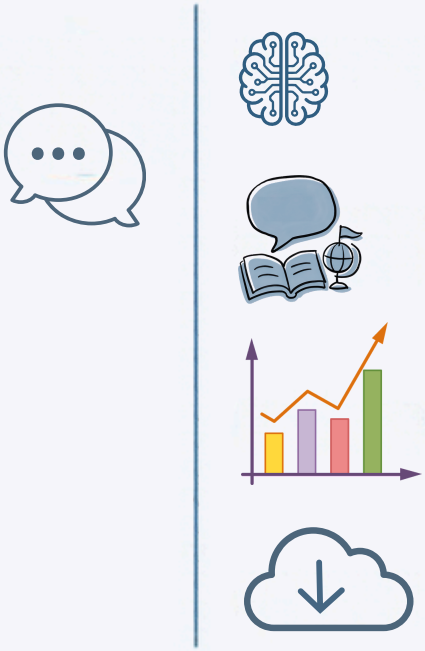
AI-Lingo

LEARN BY USING WHAT YOU ALREADY KNOW.





Project Overview

AI-Lingo is an AI-powered language-learning application designed to help learners practice language skills through personalized, real-world conversational scenarios based on their interests, learning goals, proficiency level, and previously learned vocabulary.

The application uses Prompt Engineering techniques, particularly Prompt Chaining and Constrained Prompting, to analyze available learner context, generate relevant scenarios, and produce level-appropriate instructional dialogues that encourage vocabulary reinforcement.

AI-Lingo will be developed using Flutter and Dart, applying object-oriented design principles, including interfaces and mixins where appropriate. Cubit/BLoC will manage application states and learning progress, while Hive NoSQL will store learned vocabulary, progress data, and selected lesson snippets for offline access. Appropriate encryption and the Android Keystore System will be used where applicable to protect locally stored learning data and cryptographic keys.

Objectives

01 Personalize Language Learning: Generate conversational scenarios based on learners' interests, goals, and practical learning needs.

02 Adapt Dialogue Complexity: Use available learner information and, where implemented, assessment responses to guide AI-generated dialogues toward an appropriate proficiency level.

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03 Apply Prompt Engineering: Implement Prompt Chaining and Constrained Prompting to structure AI generation and prioritize previously learned vocabulary while introducing relevant new expressions.

04 Support Real-World Practice: Provide instructional dialogues based on everyday, academic, and professional situations.

05 Track Learning Progress: Preserve learned vocabulary, progress information, and selected lesson content for continued access.

06 Ensure Maintainability and Data Protection: Apply modular object-oriented design, structured state management, local persistence, and appropriate security mechanisms.

Project Scope

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The initial version of AI-Lingo will include:

- Learner Context and Proficiency Guidance: Collect learner preferences, goals, and selected proficiency level, with optional AI-assisted level estimation based on assessment responses.
- Personalized Scenario Generation: Generate practical conversational scenarios based on learners' interests and real-world learning needs.
- AI-Generated Instructional Dialogues: Produce dialogues appropriate to the learner's available proficiency information and selected scenario.
- Vocabulary-Aware Learning: Encourage the use of previously learned vocabulary while introducing a limited number of relevant new expressions.
- Interactive Conversational Practice: Allow learners to engage with AI-generated dialogues through a text-based conversation interface.
- Learning Progress and Vocabulary Review: Enable learners to review saved vocabulary, track learning progress, and revisit selected lesson content.
- Offline Access to Saved Content: Provide access to previously saved vocabulary and selected lesson snippets without an internet connection.

The initial release will focus on English language learning through text-based conversational practice and personalized scenarios. Support for additional languages may be considered in future versions. Real-time voice conversations, pronunciation scoring, advanced adaptive learning, and fully offline AI generation will remain outside the initial scope.